

Integration of Retrieval-Augmented Generation Technique for LLM-based Differential Diagnosis Assistant

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Abstract

Artificial Intelligence (AI) is increasingly transforming the medical field, offering significant potential for diagnosis, treatment, and patient care. However, its successful integration relies on healthcare professionals, such as doctors, psychologists, and nurses, trusting the technology's reliability and accuracy. For Large Language Models (LLMs), this trust requires transparent, verifiable, and rigorously reviewed information sources. This paper presents an AI-powered tool for differential diagnosis and disease comparison, utilizing an LLM enhanced by Retrieval-Augmented Generation (RAG). RAG overcomes traditional LLM limitations by enabling access to external, domain-specific knowledge, ensuring accurate and contextually relevant responses. The system leverages PubMed, a biomedical article aggregator, to extract symptom-related information from scientific literature on various disorders. Evaluations involving psychologist-administered questionnaires demonstrate that combining a similarity score with detailed symptom descriptions provides a clear understanding of relationships between disorders. This approach may enhance diagnostic precision and build trust in AI-driven tools, encouraging their broader adoption in clinical practice.

CCS Concepts

• **Computing methodologies** → **Information extraction**; • **Information systems** → **Language models**; • **Applied computing** → **Health informatics**.

Keywords

Large Language Models, Retrieval-Augmented Generation, e-Health, Differential diagnosis

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1 Introduction

Artificial intelligence (AI) is increasingly becoming an integral part of people's daily lives, revolutionizing various sectors with its ability to process vast amounts of data, recognize patterns, and make predictions with unprecedented accuracy. The field of medicine has experienced significant transformation due to the rapid advancements in AI technologies. Indeed, AI is reshaping how healthcare is delivered and managed thanks to the introduction of innovative diagnostic tools and personalized treatment plans [3, 17, 28]. As the integration of AI in healthcare continues to evolve, it is crucial to explore its impact on the medical practice in terms of acceptance and effectiveness.

The concept of AI involves simulating intelligent behaviors with limited human intervention [8]. Today, it is widely employed to support physicians, giving rise to augmented medicine; i.e., a branch that leverages new technologies to enhance various aspects of clinical practice [2]. However, professionals often exhibit reluctance toward these technologies, partly due to insufficient digital training [7]. Distrust is further caused by a lack of transparency in the sources used, particularly in the case of Large Language Models (LLMs). LLMs are AI systems trained on vast datasets of text from articles, books, and online sources, learning linguistic patterns to perform natural language processing tasks [21]. While these models require extensive data for training, research indicates that data quality is more critical than quantity [26]. Thus, using reliable sources



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and enabling users to verify data provenance is essential. Additionally, LLMs are prone to hallucinations; i.e., instances where they generate incorrect or nonsensical information convincingly presented as factual [18]. In medicine, this poses a significant barrier to adoption, as professionals hesitate to rely on such systems for diagnostic support [22]. Despite these challenges, the potential of LLMs is evident. However, the black-box nature of these models, where training data and architecture remain undisclosed, hinders trust in their outputs. While responses are intended for review by domain experts, errors are not always easily detectable. This issue intensifies in specialized fields where specific knowledge may be absent from the training data. Fortunately, LLMs' ability to rapidly assimilate and reformulate information allows users to supplement the system with necessary knowledge [21].

AI techniques are already extensively used across various roles and fields in medicine. Recent advancements include the integration of AI in robotic surgical assistants [1] and the application of models for predicting protein complexes in research [8]. This paper proposes a novel use of AI as a diagnostic assistant. While AI tools are already employed in diagnostics, such as Machine Learning (ML) models for early tumor detection trained on large datasets of X-rays [11], these systems often gain greater acceptance due to their reliance on hospital-provided training data. However, generative AI systems face skepticism despite promising results, hindering their widespread adoption in medicine. The proposed solution aims to bridge this gap by assisting physicians during diagnosis, leveraging reliable data sources to address uncertainties and enhance decision-making. Reliable data sources fall into two categories: Electronic Medical Records (EMRs) and peer-reviewed literature. EMRs, collected by hospitals, contain sensitive patient information such as medical history, diagnoses, treatments, and test results, making them difficult to access [13]. Peer-reviewed literature, on the other hand, consists of scientifically validated articles, offering a robust foundation for research and clinical support. By utilizing scientific articles as context, the proposed system also serves as a research aid. The user interface adopts a chatbot-like structure, with knowledge constrained to the specified medical domain, ensuring focused and relevant interactions.

The work presented in this paper focuses on integrating a Retrieval-Augmented Generation (RAG) technique into a pre-trained Large Language Model (LLM) without medical fine-tuning, aiming to assist physicians in differential diagnosis. Differential diagnosis refers to the process of distinguishing between the likelihood of a disease and other conditions with similar symptoms that could account for a patient's illness [10]. RAG is an AI framework that enhances the LLM's internal knowledge by incorporating external, domain-specific information [16]. When this external knowledge is sourced from trusted repositories, the system's responses are less prone to hallucinations and more likely to gain acceptance among medical professionals. In this study, the contextual information is drawn from PubMed, a freely accessible literature search service maintained by the National Center for Biotechnology Information (NCBI) [15]. Symptom validation is ensured through cross-referencing with the International Classification of Diseases (ICD) ontology.

To assess the system's potential for user acceptance, our solution was experimentally evaluated by presenting differential diagnosis

scenarios to a group of psychologists. They evaluated responses generated by three methods: an LLM with RAG using full abstracts, ChatGPT-4, and our proposed system. Detailed results will be discussed later, but preliminary findings indicate that the majority of participants view this technology as useful, particularly for diagnostic assistance and research. In terms of comparative analysis, the proposed system outperformed the others, while ChatGPT-4 was preferred for clarity of exposition. The RAG-based method with full abstracts excelled in providing traceable sources, a critical feature for establishing trust. These insights highlight the strengths of each approach and underscore the potential of the proposed system to support medical decision-making while addressing the need for transparency and reliability.

To summarize, the main contributions of our paper are the following:

- Development of a methodology to extract symptom-focused context from PubMed, a leading biomedical literature aggregator, ensuring relevance and accuracy for diagnostic purposes.
- Integration of the Retrieval-Augmented Generation (RAG) technique into a pre-trained Large Language Model (LLM), enabling it to assist medical professionals in differential diagnosis by leveraging external, domain-specific knowledge.
- Comprehensive evaluation of the system through questionnaires administered to psychologists, comparing its performance against a knowledge-enhanced pretrained language model and ChatGPT-4o, highlighting its strengths and areas for improvement.
- Design and implementation of an intuitive graphical user interface (GUI) to facilitate seamless interaction between the system and physicians, enhancing usability and adoption in clinical settings.

The rest of the paper is structured as follows. Section 2 discusses related work. Section 3 introduces our proposed solution. Section 4 presents the architecture and implementation of our methods. Section 5 illustrates our experimental evaluation. Section 6 discusses the achieved results. Section 7 concludes the paper and highlights future developments.

2 Related work

Since the primary goal of our system is to provide diagnostic assistance, the relationships between symptoms and diseases are central to the problem. Health workers and general practitioners acquire most of their knowledge about disease symptoms through direct observation in hospitals. However, transferring this information in large quantities for use as a knowledge base remains challenging. Additionally, as discussed, EMRs present difficulties in both retrieval and maintaining patient privacy. A viable solution would be the construction of a knowledge base that systematically relates symptoms to diseases. To this aim, Zhou et al. [27] proposed the Human Symptom-Disease Network, designed to illustrate the interconnections between symptoms, diseases, genes, and proteins in the clinical manifestation of illnesses.

The advent of Generative AI techniques, particularly LLMs, brought several advantages, including rapid and structured search tool capabilities, summarizing complex information, and assisting in

diagnosis. However, significant limitations hinder the unrestricted use of LLMs in the medical domain. One of the most pressing concerns in this context is the model's strong dependence on the quality and nature of its training data, which can lead to incorrect responses or hallucinations. Furthermore, LLMs lack transparency, struggle to differentiate between reliable and unreliable sources, and operate on static, difficult-to-update data [4].

Recent advancements have demonstrated how fine-tuning pre-trained models with domain-specific data can significantly enhance performance [21]. Solutions such as Med-PaLM 2 and BioBERT exemplify this approach. Med-PaLM 2, an LLM optimized for medical question answering, has achieved state-of-the-art results across multiple MultiMedQA benchmarks, including MedQA multiple-choice questions [19]. BioBERT, on the other hand, is a domain-specific language model pre-trained on large biomedical corpora (PubMed and PubMed Central) to enhance biomedical text mining tasks such as named entity recognition, relation extraction, and question answering [12]. The work proposed in this paper addresses these last two issues, but follows a different approach.

In recent years, Retrieval-Augmented Generation (RAG) has emerged as a solution to mitigate the limitations of LLMs, such as hallucinations, lack of transparency, and accuracy concerns. RAG integrates the inherent knowledge of LLMs with external databases, making it particularly effective in solving real-world problems that require domain-specific expertise [6]. This capability has garnered interest in the medical field, where reliability and accuracy are paramount. Studies evaluating RAG-based systems in medical question-answering tasks indicate that, depending on the query, RAG often surpasses traditional LLMs while maintaining consistency with state-of-the-art models [25]. A key advantage of RAG with respect to fine-tuning is that the former does not require extensive computational resources [6].

Even with a comprehensive literature base such as PubMed, there is no guarantee that sufficient information is available for every disease. This challenge is further compounded when investigating relationships between diseases, making it less likely to find a large number of relevant articles. Fortunately, research indicates that RAG can achieve superior results even with limited initial data, outperforming models based solely on fine-tuning [20]. Additionally, studies suggest that smaller models enhanced with external knowledge can outperform larger models.

In this work, we selected LLM from the Llama 3 family developed by Meta [5]. Specifically, we used Llama 3-Instruct, a fine-tuned version optimized for instruction-following tasks. This open-source model is available in different configurations, including 8 billion and 70 billion parameters. After initial training, it undergoes supervised fine-tuning followed by reinforcement learning with human feedback (RLHF). Special attention has been given to ensuring the model's safety, preventing the generation of harmful, biased, or unreliable content. This is particularly evident in its handling of medical queries, where the model often advises users to consult healthcare professionals. Llama's combination of scalability, competitive performance, and open-source accessibility has made it an attractive foundation for domain-specific medical modeling. For instance, projects such as PMC-LLaMA [24] leverage extensive biomedical corpora to enhance the model's expertise in the medical field. The authors of PMC-LLaMA are also responsible for MedC-K,

a large academic paper aggregator, which is essential for training knowledge-intensive models. Additionally, an instruction-tuning process is applied, refining the model using a collection of tasks structured as instructions. The primary motivation behind this project is to develop an open-source alternative with performance comparable to GPT-3.5 while retaining the flexibility and adaptability of Llama models. Future work suggests that integrating RAG will be crucial to mitigating the hallucination problem inherent to LLMs.

3 Proposed solution

In this section, we introduce the requirements of our system, and present the main components of our solution.

3.1 Requirements

A fundamental requirement of our diagnostic support system is reliability. This is ensured by leveraging authoritative knowledge sources and models trained on such information, minimizing reliance on independent reasoning. PubMed serves as an excellent repository of biomedical abstracts. While not all articles are peer-reviewed, 91% of its content is indexed in MEDLINE, the primary bibliographic database of the National Library of Medicine (NLM) [23]. PubMed's advantages include free access, broad article availability, and multiple interface options. In addition to being accessible via its website, it can be queried through the public E-utilities API, provided by the Entrez system of the National Center for Biotechnology Information (NCBI), which grants access to all Entrez databases, including PubMed. Another key benefit is its constantly updated and downloadable article database.

To validate the information extracted from PubMed, we use the International Classification of Diseases (ICD) ontology. The ICD provides a standardized language for recording, reporting, and monitoring diseases. It serves as a reference for health records and statistical data across primary, secondary, and tertiary care. Another crucial requirement is minimizing hallucinations and misinformation in the use of pre-trained models. To mitigate such risks, a detailed prompt is employed during the symptom extraction phase, followed by rigorous verification to ensure that only symptoms or related diseases are reported. Distinguishing between symptoms and diseases is inherently challenging, as the boundary between them is often ambiguous.

3.2 Rationale of our solution

The phenomenon of hallucinations in large language models (LLMs) poses a significant challenge, often stemming from the general nature of their training data, which may lack domain specificity. There are two main solutions to this problem. One solution is *fine-tuning*; i.e., retraining the model on a specialized subset of data to address a particular problem. Another solution is *Retrieval-Augmented Generation* (RAG). The latter uses non-parametric memory (i.e., data not incorporated during training) stored as a dense vector index accessible via a pre-trained retriever [14]. The input query vector is compared with stored document vectors, retrieving the top-K most relevant documents. These are then integrated into the prompt of the pre-trained model, supplementing its parametric memory with relevant knowledge for response generation. The key components

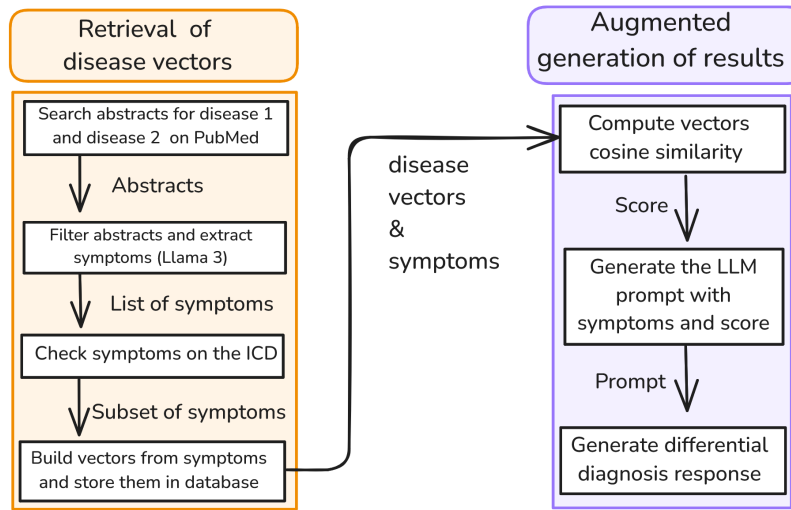


Figure 1: A graphical representation of the RAG technique, from the extraction of the knowledge base to the LLM response.

of a RAG system include a pre-trained model, a retriever, and a vector database.

The advantages and disadvantages of RAG compared to fine-tuning are as follows:

- **Knowledge dynamism:** RAG is preferable when information is frequently updated, as fine-tuning requires retraining to integrate new data.
- **Source data:** fine-tuning necessitates large, annotated datasets, whereas RAG only requires domain-specific knowledge. However, acquiring this knowledge can be time-consuming or, in some cases, infeasible.
- **Computational cost:** RAG eliminates the need for retraining, reducing training costs. However, retrieval-based response generation may become computationally expensive, particularly with large document chunks.
- **Transparency:** RAG allows direct inspection of retrieval sources, ensuring traceability of information. Fine-tuned models may generate responses without explicit source attribution.
- **Flexibility and reasoning:** Fine-tuned models benefit from deeper domain familiarity, enabling feature extraction, reasoning, and hypothesis generation. Classical RAG remains rigid, strictly adhering to retrieved context.

The decision to adopt RAG in this research is driven by the strict need for reliable, verifiable sources. Fine-tuning is computationally intensive, requiring significant hardware resources, making it less viable in constrained environments. Furthermore, data sharing restrictions in the medical field complicate the availability of large, annotated datasets for fine-tuning.

To focus on the most critical aspect of differential diagnosis, we extract only symptoms from texts instead of full abstracts. Indeed, in our RAG application, using an extensive concatenation of abstracts extracted from scientific papers as context would be computationally expensive and may dilute relevance. Differential diagnosis hinges on shared and distinguishing symptoms; thus, in our RAG implementation, diseases are represented as vectors of

associated symptoms. The similarity between two disease vectors quantifies their symptom overlap.

The retrieval phase, illustrated in Figure 1, constructs input disease representations as vectors of closely related symptoms and diseases. These vectors are stored in a non-relational database and retrieved for comparison when queried. PubMed serves as the primary source of abstracts, extracted using an LLM with a dedicated prompt for symptom identification. Extracted terms undergo validation by checking for at least one matching node in the International Classification of Diseases (ICD) with a similarity score above 0.4. Valid symptoms are incorporated into disease vectors, which are stored and later used during augmented generation. Additionally, cosine similarity between disease vectors quantifies their difference, contributing to the model’s external knowledge.

4 Architecture

This section presents our system’s architecture, highlighting the technologies used to achieve the system’s goals. It also describes user interaction with the system, covering all necessary steps from input to response generation, including the system’s internal processes.

4.1 Overview

In the following, we summarize the key system steps. Users specify the diseases they wish to analyze, forming the basis for context construction using abstracts from PubMed. To retrieve abstracts without downloading the database locally, the system uses the NCBI Entrez E-utilities API, interfaced via the metapub library, which provides Python objects representing PubMed papers. The ICD search function relies on its HTTP-based REST API, which requires authentication.

For abstract analysis and symptom extraction, the system employs LLaMA 3 Instruct (8B parameters). LLaMA 3 follows a standard Transformer architecture, with improvements in training data quality and volume [5]. It is configured with a temperature of 0.6

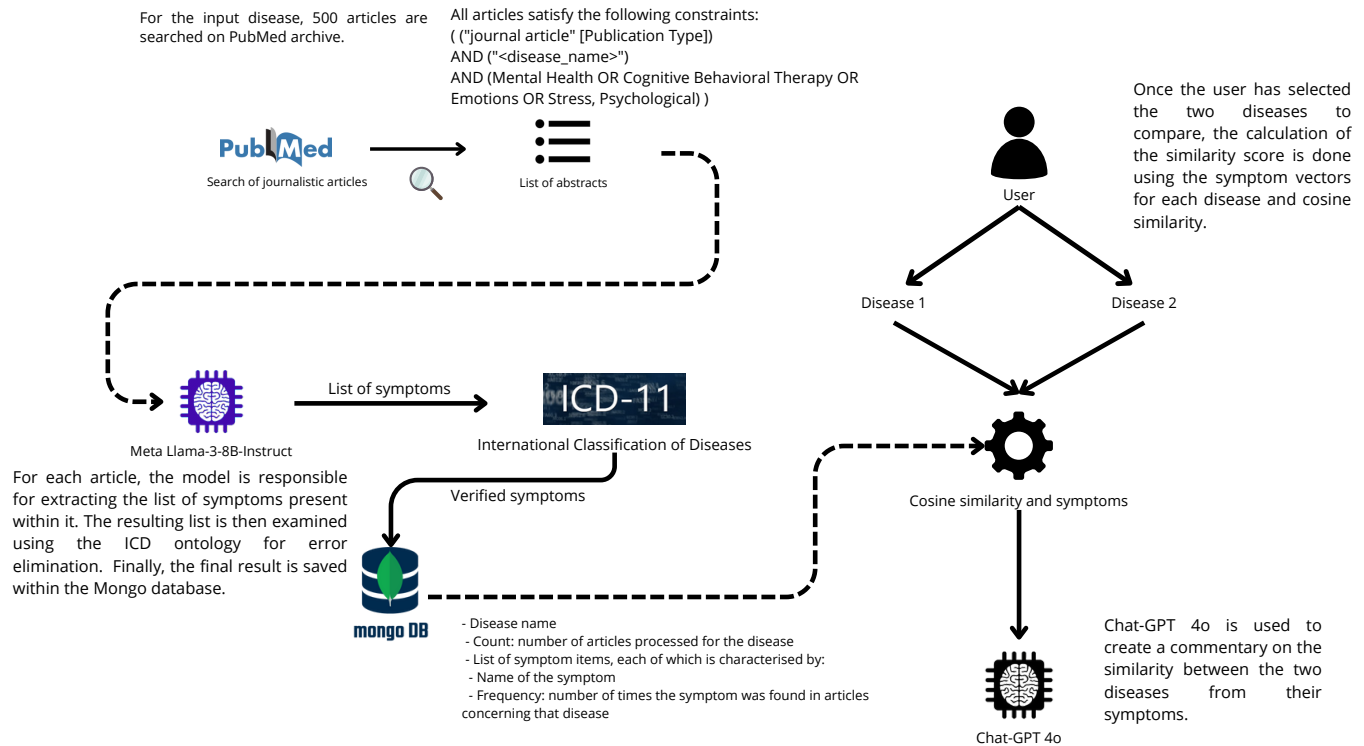


Figure 2: System workflow: disease vector creation, storage, and user interaction

and a top-p sampling of 0.9, ensuring robust instruction-following capabilities suited for medical applications.

Extracted disease vectors are stored in MongoDB Atlas, a NoSQL database. Each entry consists of:

- Disease name;
- Number of abstracts used;
- List of symptom elements, each characterized by its name and frequency (i.e., occurrence count in abstracts).

A higher frequency typically indicates a symptom's significance for the disease. Given the simple queries and structured nature of the data, a NoSQL database was preferred over relational alternatives, improving retrieval efficiency and response times.

4.2 Workflow

Below, we outline the system's workflow. At first, the user must specify diseases of interest. Pre-computing disease vectors ensures efficiency, as real-time extraction and symptom identification would slow down the system. Figure 2 illustrates the complete workflow for Bipolar Disorder, detailing the process from vector creation to user interaction.

- (1) **Context construction:** Before comparison, disease vectors must be built. Disease names serve as search terms in PubMed via metapub. The system retrieves up to 500 abstracts per disease, filtering them based on the following constraints:
 - Published after 2020
 - Classified as journal articles

- Contain the disease name and at least one relevant keyword

Keywords focus on psychological disorders: *Mental Health, Cognitive Behavioral Therapy, Emotions, Stress, Psychological*. The system extracts the title, abstract, and publication year from selected articles.

- (2) **Symptom extraction:** The abstracts are processed using LLaMA 3 Instruct (8B). A detailed prompt specifies the source (PubMed), provides few-shot examples, describes the extraction task, defines output format constraints, and includes the relevant abstracts.
- (3) **ICD validation:** Extracted terms are validated against the ICD via API. The system searches for the term, assessing all returned nodes. Only nodes with a similarity score above 0.4 are considered valid symptoms and added to the disease vector.
- (4) **Database storage:** Finalized disease vectors are stored in MongoDB Atlas. The database allows dynamic updates: new diseases can be added, symptoms modified, and records removed.
- (5) **Comparison and similarity calculation:** When a physician selects two diseases, their corresponding vectors are retrieved and compared using cosine similarity, quantifying symptom overlap. If a comparison exists in the database, it is reused to save computation time.
- (6) **Response generation:** The similarity score, along with disease names and symptoms, is formatted into a prompt for

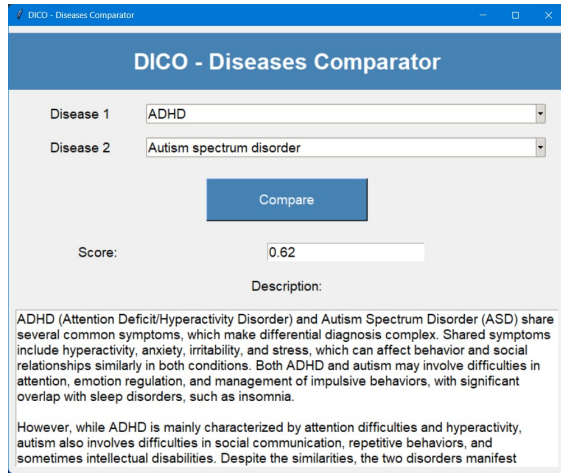


Figure 3: GUI for physician interaction with the system

the final LLM (ChatGPT 4o), which generates a user-friendly explanation:

Your task is to provide a textual description of the similarities between these two diseases based on a list of symptoms. The total similarity score for these two diseases is {score}. The diseases are {disease_1} and {disease_2}. Below are the symptoms for each disease: {symptoms_disease_1} {symptoms_disease_2}.

- (7) **User interface and query efficiency:** The response is stored in MongoDB Atlas for fast retrieval in future queries. A graphical user interface (GUI) facilitates seamless interaction.

Figure 3 showcases the GUI, where physicians can select disorders for comparison, view similarity scores, and access detailed explanations. Average response time is approximately 5 seconds for new comparisons, slower than standard ChatGPT 4o due to data retrieval and validation steps.

5 Experimental evaluation

To assess the effectiveness of our system, we conducted an experiment with experts evaluations. A group of 12 professional psychologists served as subject matter experts and completed 24 questionnaires, each comparing three different response examples on a couple of disorders. The written responses were generated by three distinct systems, denoted as A, B, and C. The participants were blinded to the system correspondence. System A (Our system) represents the approach proposed in this paper, System B (ChatGPT) corresponds to direct ChatGPT-4o outputs, and System C (Abstracts) is an alternative retrieval-augmented generation (RAG) approach. In System C, instead of incorporating all symptoms of the two diseases, the additional knowledge consists of abstracts from PubMed that describe the relationship between the diseases under investigation.

The evaluation process was structured as follows: For each disease pair, the three systems assigned a similarity rating along with a corresponding explanatory comment. Psychologists were then

Survey questions

- Q1: What similarity score would you assign to this disease pair?
- Q2: The similarity score calculated by the system is accurate.
- Q3: The description contains all relevant information needed to accurately distinguish the two diseases.
- Q4: The descriptions are written in clear and easily understandable language.
- Q5: The description is consistent and free of contradictions or ambiguities.
- Q6: The descriptions highlight the most clinically relevant aspects (e.g., distinctive symptoms and overlapping features).
- Q7: The details provided are useful for comparing diseases from a clinical perspective.

Table 1: Survey questions used in the evaluation. Respondents rated each statement using a five-point Likert scale: 5 = Very much agree, 4 = Agree, 3 = Undecided, 2 = Disagree, 1 = Not at all agree. For Q1, responses range from 1 (not at all similar) to 10 (synonyms).

asked to provide their own similarity rating for the disease pair, assess the system-generated ratings, and evaluate the usefulness of the accompanying comments.

Following each evaluation, participants completed a questionnaire indicating their level of agreement with a series of statements. Responses were recorded using a five-point Likert scale: *Very much agree*, *Agree*, *Undecided*, *Disagree*, and *Not at all agree*. Additionally, after completing all three system evaluations for a given disease pair, participants completed a fourth questionnaire, in which they provided feedback on their perceived usefulness.

To assess the reliability of the responses, Krippendorff’s Alpha-Reliability was used by converting the Likert-scale choices into numerical values (5 = Very much agree, ..., 1 = Not at all agree). Krippendorff’s alpha (α) is a reliability coefficient that measures the level of agreement among raters and is calculated as:

$$\alpha = 1 - \frac{D_o}{D_e}$$

where D_o represents the observed disagreement among raters, and D_e represents the expected disagreement if ratings were assigned at random [9]. The coefficient ranges from -1 to 1, with values closer to zero indicating random agreement among raters and 1 represents perfect agreement. The resulting Krippendorff’s alpha value is **0.313**, indicating moderate agreement.

As shown in Table 1, one of the key evaluation criteria is the difference between the system-assigned score and the psychologist-assigned score. To determine which system aligns best with human judgment for question Q1, the Average Grade Difference (AGD) (i.e., absolute difference between system-assigned and psychologist-assigned scores) was computed for each method. These values were

	AGD	RMSE	Q2	Q3	Q4	Q5	Q6	Q7
Our system	1.45	1.79	3.08	2.91	3.96	3.71	3.08	2.83
ChatGPT	2.00	2.45	3.25	2.96	4.00	3.92	3.37	3.08
Abstracts	1.95	2.30	3.21	3.09	3.62	3.54	3.21	3.08

Table 2: Average ratings for each evaluation criterion. AGD: (Average Grade Difference) measures the discrepancy between system-assigned similarity scores and human ratings, with lower values indicating better alignment. RMSE: (Root Mean Square Error) assesses the magnitude of error, where lower values signify closer agreement with human ratings.

averaged per method and are presented in the AGD column of Table 2. Additionally, Root Mean Square Error (RMSE) was calculated to identify potential outliers in similarity scoring. Regarding similarity score assignment, results show that our system outperforms the other ones, as it produces the smallest discrepancy with respect to psychologists ratings. All three systems demonstrate the ability to broadly distinguish related disorders, particularly when symptoms overlap. However, ChatGPT and Abstract methods struggle to capture finer distinctions necessary for precise similarity scoring. Our system’s improved performance can be attributed to its use of entity recognition techniques, which isolate and compare symptoms between disorders. To account for potential outliers, RMSE values were computed, indicating that while all three systems maintain a reasonable level of consistency, our system exhibits the lowest RMSE, suggesting fewer extreme errors in scoring.

Beyond similarity ratings, the survey also explored the perceived usefulness and applicability of the system. Table 2 presents the average ratings for each evaluation criterion. The values contributing to these averages correspond to the numerical conversion of responses, using the same scale as in the Krippendorff’s Alpha calculation. For each method, scores were aggregated and averaged across the 24 questionnaires. The evaluation focuses on questions 2 to 7, as question 1 pertains to similarity score assignment by the participants.

Across all methods, descriptions were generally rated as clear and free of contradictions. Notably, ChatGPT scored particularly well in clarity, underscoring the strength of large language models in generating comprehensible responses. However, skepticism remained regarding content completeness and usefulness in differential diagnosis. Questions 3 and 7 recorded the lowest overall averages, indicating room for improvement in providing fully comprehensive and clinically relevant comparisons. While the Abstract method provided informative content, its structured presentation was perceived as less immediate compared to ChatGPT’s conversational-style responses. In the final questionnaire, respondents were asked for their general opinion on the utility of such an approach. The majority of participants expressed agreement with its usefulness, with an average rating of 3.41 on the Likert scale.

6 Discussion

In this section, we summarize the results achieved, discuss the system’s current limitations, and propose potential solutions for future development.

6.1 Achievements

A key challenge in integrating artificial intelligence, particularly generative models, into medical decision-making is the skepticism surrounding their reliability. This skepticism primarily arises from concerns about non-authoritative information sources and the tendency of large language models (LLMs) to overgeneralize. In this study, we introduced a solution leveraging Retrieval-Augmented Generation (RAG) to address these challenges. The key achievements of our system are as follows:

- The model accurately responds to questions regarding the differential diagnosis of disorder pairs. Specifically, it identifies and justifies symptom presence based on literature aggregated from PubMed. Additionally, it calculates a similarity score to quantify the relationship between diseases. Users interact with the model through a graphical interface.
- Unlike standard generative LLMs, the system does not generate ad hoc answers when information is insufficient. Instead, it recognizes the lack of reliable sources and refrains from speculation, resulting in significantly fewer hallucinations.
- An automated pipeline was developed to create context from PubMed abstracts. This pipeline retrieves full-text discussions on the disorders and identifies relevant articles that support or challenge their connection.
- Expert evaluation of the system indicates that it is generally useful for understanding disease similarities. Additionally, findings suggest potential applications in diagnostics and research, where structured knowledge retrieval can assist clinical reasoning.

The integration of RAG into LLMs, utilizing knowledge from peer-reviewed literature, offers a promising avenue for clinical applications. While this approach mitigates the reliability issues seen in widely used generative models, whether it is sufficient to gain full confidence among physicians remains an open question. Evaluation results suggest that the system represents a step forward compared to existing solutions, though challenges remain.

6.2 Limitations

Despite its promising results, the current version of the system has several limitations that need to be addressed for further improvement. The effectiveness of a RAG system largely depends on the quality and scope of its data. To be truly valuable, the system must provide insights beyond what domain experts already know. However, our evaluation indicates that relying solely on widely cited articles may not sufficiently enhance physicians’ knowledge. At the same time, incorporating too broad a context can reduce the precision of extracted information.

Another limitation is the exclusive reliance on symptoms as context. While symptoms provide useful information, they may lack the depth required for fully informed decision-making. The system also exhibits slower response times compared to standard ChatGPT-4o. When contextual information is limited, the model produces shorter and less comprehensive responses. Additionally, if required context is not readily available, generating it from scratch adds further delays, particularly due to API constraints in document retrieval. Addressing these limitations will be crucial for making the system more practical for real-world clinical applications. They can

be addressed by optimizing retrieval efficiency, improving context selection mechanisms, and exploring complementary reasoning technologies to enhance diagnostic reliability.

7 Conclusion and future work

While the current approach shows promise, the primary challenge remains the quality and comprehensiveness of the data. Addressing this requires expanding the context used for retrieval, incorporating a structured integration of PubMed abstracts (from which symptoms are extracted) along with other relevant information, potentially including full-text articles. Although electronic medical records (EMRs) could further enhance the system, privacy constraints make their integration questionable in the near future.

Further improvements can be made in the integration of the RAG method, particularly in the retrieval phase. More advanced techniques, such as reranking documents (or symptoms) by assigning importance values, could prioritize the most relevant information, reducing the risk of diluting key details in broader contexts. Another valuable approach involves query rewriting to refine searches in the vector database, ensuring that embeddings capture relevant similarities rather than superficial aspects of the query.

A promising direction for improvement is the combination of RAG with fine-tuning. While the current model has been fine-tuned for dialogue-based tasks, it lacks domain-specific optimization for medical applications. At present, a rigid prompt structure restricts inference-making to prevent hallucinations. By leveraging fine-tuned models such as Med-PaLM 2 and BioBERT, the system could be enhanced to draw more insightful, non-obvious connections while maintaining factual accuracy. A hybrid approach that integrates parameterized expert memory with disorder-specific non-parametric knowledge could further refine the diagnostic process.

Finally, additional evaluations should be conducted with a larger and more diverse pool of domain experts. Future studies should enable participants to engage in full diagnostic assistance scenarios, allowing deeper interaction with the system and a more comprehensive assessment of its effectiveness in clinical decision-making.

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